



Lithium-ion battery fundamentals and exploration of cathode materials: A review

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ABSTRACT

Advances in cathode materials continue to drive the development of safer, more efficient, and sustainable lithium-ion (Li-ion) batteries for various applications, including electric vehicles (EVs) and grid storage. This review article offers insights into key elements—lithium, nickel, manganese, cobalt, and aluminium—within modern battery technology, focusing on their roles and significance in Li-ion batteries. The review paper delves into the materials comprising a Li-ion battery cell, including the cathode, anode, current concentrators, binders, additives, electrolyte, separator, and cell casing, elucidating their roles and characteristics. Additionally, it examines various cathode materials crucial to the performance and safety of Li-ion batteries, such as spinels, lithium metal oxides, and olivines, presenting their distinct advantages and challenges for battery applications. Lithium manganese (Li-Mn-O) spinels, like LiMn_2O_4 , offer a cost-effective and environmentally friendly option with good thermal stability despite challenges such as capacity fading, which necessitate innovative approaches like dual-doping strategies. Nickel-rich lithium metal oxides like $\text{LiNi}_x\text{Mn}_y\text{Co}_{1-x-y}\text{O}_2$ provide high specific energy but face/encounter issues with cobalt reliance and stability, prompting research to reduce cobalt content and increase nickel content. Olivine-based cathode materials, such as lithium iron phosphate (LiFePO_4), prioritize safety and stability but exhibit lower energy density, leading to exploration into isomorphous substitutions and nanostructuring to enhance performance. Safety considerations, including thermal management and rigorous testing protocols, are essential to mitigate risks of thermal runaway and short circuits. Thus, this review scrutinizes recent advancements in Li-ion battery cathode materials, delving into strategies aimed at mitigating associated drawbacks and identifying suitable electrode materials based on their electrochemical performance and capacity during operation.

1. Introduction

The introduction and subsequent commercialization of the rechargeable lithium-ion (Li-ion) battery in the 1990s marked a significant transformation in modern society. This innovation quickly replaced early battery technologies, including nickel zinc, nickel-metal-hydride, and nickel-cadmium batteries (Batsa Tetteh et al., 2022). In contrast to its predecessors, Li-ion batteries exhibited superior attributes, such as high energy and power density, extended lifespan, minimal self-discharge, and a relatively low environmental impact (Batsa Tetteh et al., 2022). These distinctive characteristics have played a pivotal role in advancing the quality of portable electronic devices such

as smartphones and tablets, a trend that is now extending to electric vehicles (EVs) (Deng, 2015). This transition is further underscored by recent technological advancements, which have prompted widespread reconsideration of EV adoption and integration. Consequently, Li-ion batteries are increasingly being recognized for their versatility and reliability, leading to a surge in their applications, particularly within the EV sector. The exponential growth of operational EVs over recent years attests to this phenomenon, with numbers escalating from under 1 million in 2014 to surpassing 11 million globally by 2022 (Jung et al., 2023). This remarkable growth trajectory not only reflects the increasing viability and acceptance of EV technology but also signifies a profound shift toward sustainable transportation solutions on a global

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scale.

It is anticipated that within the next decade, EVs will surpass the extensive use of internal combustion engine (ICE) vehicles, playing a critical role in significantly reducing carbon pollution, particularly in the road transport sector (Rangarajan et al., 2022, Roy et al., 2022). This increase can be attributed to a heightened demand for eco-friendly transportation alternatives, coupled with increased support from both governmental and private sectors (Poon, 2022). However, the widespread adoption of battery-based EVs in the transportation sector, primarily reliant on Li-ion batteries, faces significant challenges stemming from key limiting factors that demand urgent attention. These challenges include the need for lightweight construction, increased energy and power density, faster charging capabilities, extended lifespan, optimal cyclability, improved safety measures, and enhanced cost-effectiveness. Addressing these issues is crucial to matching or surpassing the performance of ICEs found in conventional vehicles (Rangarajan et al., 2022, Nzereogu et al., 2022, Alanazi, 2023). Although EVs are often promoted as vehicles with zero tailpipe emissions, their transition from ICEs is impeded by these challenges, which are central to long-term decarbonization strategies for the transportation sector (Malima and Moyo, 2023).

At the core of the Li-ion battery lies an intricate electrochemical arrangement crucial for its functionality. In Li-ion rechargeable batteries, the cathode plays a vital role by storing lithium ions through electrochemical intercalation, requiring adequate lattice sites or voids to enable the reversible storage and release of active ions. During charge and discharge cycles, lithium ions are reversibly introduced into and removed from the cathode and anode materials (Goodenough and Park, 2013, Ün, 2023). Thus, robust crystal structures with sufficient storage sites are imperative for producing materials with consistent cycling stability and a high specific capacity. Additionally, a cathode with a high potential for electrochemical intercalation can contribute to a battery with high energy density when paired with a specific anode. This is because the energy density of the battery is a function of the electrode materials specific capacities and the operating voltage, which is significantly influenced by the electrochemical potential differences between the cathode and anode (Liu et al., 2016, Kaur and Gates, 2022, Yusuf, 2021).

Li-ion batteries come in various compositions, with lithium-cobalt oxide (LCO), lithium-manganese oxide (LMO), lithium-iron-phosphate (LFP), lithium-nickel-manganese-cobalt oxide (NMC), and lithium-nickel-cobalt-aluminium oxide (NCA) being among the most common. Graphite and its derivatives are currently the predominant materials for the anode. The chemical compositions of these batteries rely heavily on key minerals such as lithium, cobalt, manganese, nickel, and aluminium for the positive electrode, and materials like carbon and silicon for the anode (Goldman et al., 2019, Zhang and Azimi, 2022). These materials are fundamental to efficient energy storage and release within the battery cell (Liu et al., 2016, Cabello et al., 2017). Therefore, the continual development of electrodes is a critical aspect of advancing high-performance EV batteries (Ju et al., 2023).

Electrolytes, separators, and current collectors facilitate ion movement between the two electrodes, directly influencing the battery efficiency and overall functionality. Factors such as electrochemical kinetics, ionic conductivity, and the durability of these components are intricately linked to the performance of Li-ion batteries (Abubaker et al., 2024). This review focuses on Li-ion batteries and the diverse materials that constitute their components, emphasizing both the challenges and innovative solutions. It examines recent advancements in battery technologies, highlighting conventional materials and their associated challenges, while also exploring emerging technologies and future developments across various battery components. Additionally, the review categorizes cathode materials based on their chemistries and technologies, aiming to identify suitable electrode candidates based on electrochemical performance and capacity. Furthermore, it addresses the advancements, advantages, limitations, and weaknesses of these

cathode materials, offering insights into the current and future state of Li-ion battery technology.

2. Overview of key elements in modern battery technology

Lithium, a key component of modern battery technology, serves as the electrolyte's core, facilitating the smooth flow of ions between the anode and cathode. Its lightweight nature, combined with exceptional electrochemical characteristics, makes it indispensable for achieving high energy density (Nzereogu et al., 2022). The remarkably low standard reduction potential of lithium, measured at -3.05 V at 298 K, allows for the production of an extremely high capacity of 3860 mA h g⁻¹. This capacity significantly surpasses alternative metals used in batteries; sodium yields only 1160 mA h g⁻¹, and zinc offers 820 mA h g⁻¹. Such a characteristic makes lithium highly desirable in the fabrication of high-density and high-voltage battery cells (Varzi et al., 2020, Mondal and Das, 2022). However, due to its high reactivity, precautions must be taken to shield lithium from exposure to both air and water to prevent potentially hazardous reactions. Asymmetric lithium battery systems require secure and tamper-resistant sealing to prevent both accidental and intentional tampering. These systems also use organic electrolytes instead of aqueous ones to mitigate lithium's reactivity (Mondal and Das, 2022). According to Theodore (2023), non-aqueous electrolyte solutions, carefully prepared and validated by researchers in the field of lithium electrochemistry a few years earlier, were employed in the development of a Li-ion battery. This battery exhibits a higher cell potential compared to the lead-acid battery, measuring 2.5 V as opposed to the latter's 2 V. This breakthrough marked a significant advancement in battery technology, paving the way for the development of more efficient and powerful energy storage systems (Theodore, 2023).

Nickel, known for its high energy density, plays a crucial role in positive electrodes, allowing batteries to store more energy and enabling longer travel ranges between charges—a significant challenge in widespread EV adoption (Lu et al., 2022). Cathodes with high nickel content are of great interest to researchers and battery manufacturers, as they are required for achieving the desired specific energy and energy density. This enhanced energy density improves performance metrics, extending the driving range and battery life of EVs. By utilizing nickel, these benefits are attainable at a lower cost, which is a major advantage given that the cost of EV batteries has been a significant deterrent to market acceptance (Zeng et al., 2018). Consequently, manufacturers are incorporating higher amounts of nickel than in previous instances, indicating a trend toward maximizing these advantages (Purwanto et al., 2022). The early generation of NMC batteries, which presently dominate cathode chemistry, initially included approximately 33 % nickel. Subsequent generations have progressively increased the nickel content, such as in the case of NMC 811, which contains 80 % nickel, and the latest generation of NMC batteries, featuring a 90 % nickel cathode (Purwanto et al., 2022, Ghosh et al., 2021). This trend underscores a continuous effort to maximize nickel content for enhanced battery performance and efficiency.

While the role of nickel in boosting energy density and efficiency is significant, manganese is equally crucial in battery performance due to its ability to provide outstanding rate capability, allowing for high current discharge while maintaining low temperatures. This feature not only enhances the safety and stability of the battery but also ensures extreme durability across numerous cycles, despite the relatively low specific energy of manganese (Koech et al., 2024). Moreover, the presence of manganese in the cathode material reduces the risk of thermal runaway—a critical safety concern—by lowering the exothermic reaction during overcharging or short-circuiting (Yang et al., 2023). Additionally, manganese reinforces the electrode's structural stability, which is vital for the battery's longevity and reliability (Schipper et al., 2017), even though it has a lower specific strength (Satpathy and Pamuru, 2021). Its ability to mitigate the degradation of layered structures during cycling, thereby reducing capacity fading, further contributes to

extending the lifespan of the battery (Xu et al., 2012).

In NMC cathode materials, cobalt plays a crucial role in enhancing thermal stability by maintaining the structural integrity and safety of the battery under high-temperature conditions, thereby reducing the risk of thermal runaway. Additionally, cobalt improves low-temperature performance, enabling efficient operation in colder climates, and increases the initial discharge capacity, leading to higher energy density and longer usage times between charges. Cobalt also optimizes rate performance, allowing for faster charging and discharging without compromising overall performance. These attributes make cobalt particularly valuable for EVs and portable electronic devices. However, NMC cathodes containing cobalt have faced scrutiny due to environmental and ethical implications in mining (Chu et al., 2022, Cunningham, 2022). In response, there is a concerted effort to reduce or replace cobalt in battery formulations, promoting more sustainable alternatives (Chu et al., 2022). Despite these challenges, the combination of manganese, nickel, and cobalt in NMC chemistry creates a stable, high-specific-energy configuration. This battery system leverages the unique advantages of each element: manganese contributes to stability, nickel enhances capacity, and cobalt improves energy density. Together, these elements provide several benefits, including relatively high capacity, the ability to handle high loading, and cost savings, as the battery does not require complex built-in circuits (Koech et al., 2024). The aluminium found in NCA chemistries provides greater stability, resulting in long service life and high specific energy, comparable to NMC, while maintaining good specific power strength (Satpathy and Pamuru, 2021). The typical ratio of nickel, cobalt, and aluminum in NCA is 8:1.5:0.5, with aluminum constituting a very small proportion that may vary to a ratio of 8:1:1. This makes NCA compositionally similar to binary materials. Therefore, the amount of aluminum in NCA typically varies from 5 % to 10 % (Lebens-Higgins et al., 2019, Julien and Mauger, 2020). The Table 1 below summarizes the properties, benefits, and limitations of lithium, nickel, manganese, cobalt, and aluminum, which are key components in Li-ion batteries.

3. Lithium battery components and functionality

Typically, a basic Li-ion cell (Fig. 1) consists of a positive electrode (the cathode) and a negative electrode (the anode) in contact with an electrolyte containing Li-ions, which flow through a separator positioned between the two electrodes, collectively forming an integral part of the structure and function of the cell (Mosa and Aparicio, 2018). Current collectors, commonly constructed of aluminium for the cathode and copper for the anode, are selected for their high electrical conductivity. These collectors, essentially metallic foils, function as conductive materials, collecting, concentrating, and distributing electrical current

during battery operation. Additionally, they serve as terminals for connecting individual battery cells externally, thereby facilitating the flow of electric current to and from the battery (Lamiel et al., 2022, Khatibi et al., 2022).

3.1. Anode and cathode dynamics

The anode and cathode electrodes play a crucial role in temporarily binding and releasing lithium ions, and their chemical characteristics and compositions significantly impact the properties of a lithium-ion cell, including energy density and capacity, among others. The anode typically consists of a graphite-based slurry layered onto the copper foil current collector, while the cathode is often composed of transitional metal oxides such as LCO, LMO, NMC, NCA or phosphate slurry-LFP coated onto an aluminium foil current collector, as highlighted by Guo et al. (2021) and Zhang et al. (2023). Recent research focuses on improving the performance of cathodes such as LiNiMnCoO₂ or LiNi-CoAlO₂ (Chen et al., 2022).

The transition metals (such as cobalt, nickel, manganese, etc.) used in cathode development can make up to 14 % of the battery mass and significantly influence the cost-effectiveness of battery recycling, accounting for 51 % of the recycling cost (Gao et al., 2015). Additionally, they are the primary contributors to eco-toxicological biohazards, especially in waste management, energy-intensive supply chains, and anticipated market expansion. This consideration is underscored by rapidly changing regulations that increasingly mandate the use of recycled metals in the production of new batteries (Muller et al., 2021).

Noteworthy anode materials, including alloy types (i.e., tin, silicon, zinc, and Germanium-based), carbon-based (i.e., graphite, graphene, carbon nanofibers, carbon nanotubes), transition metal oxides, and silicon composites, present distinct advantages and limitation (Hossain et al., 2023). Silicon composites offer an enhanced capability range of 3579 mA h g⁻¹ but are prone to sudden charge-induced volume expansion of up to 300 % and unreliable solid electrolyte interphase (SEI) formation (Szczec and Jin, 2010, Guo et al., 2019), which exert significant strain on the electrode material. This strain may lead to cracking and fragmentation, ultimately reducing conductivity (Guo et al., 2019). Lithium metal anodes are distinguished by their superior energy density compared to other anode materials, making them a promising technology. However, despite this potential, they face significant challenges, including dendrite formation, electrolyte instability, and safety concerns, which hinder the full exploitation of lithium's benefits in battery technology (Liu et al., 2019). Transition metal oxides (TMOs), including selenides, sulfides, nitrides, and phosphides, are used as conversion-type anode materials due to their high specific capacity (600–1000 mA h g⁻¹) and considerable stability. However, they are

Table 1
Comparative properties and performance characteristics of key battery metals.

Metal	Properties	Benefits	Limitations	Citations
Lithium	Core component of the electrolyte, low standard reduction potential (-3.05 V), high capacity (3860 mA h g ⁻¹).	High energy density, lightweight, excellent electrochemical characteristics, facilitates high-capacity batteries.	Highly reactive; requires protection from air and water, needs secure and tamper-resistant sealing.	(Nzereogu et al., 2022, Varzi et al., 2020, Mondal and Das, 2022, Theodore, 2023)
Nickel	High energy density, improves battery performance, commonly used in NMC cathodes with varying nickel content (33 % to 90 %).	Increases energy density and battery efficiency, extends driving range of EVs, lower cost compared to some alternatives.	Stability issues at high temperatures, higher cost than some other metals.	(Lu et al., 2022, Zeng et al., 2018, Purwanto et al., 2022, Ghosh et al., 2021)
Manganese	Provides high-rate capability, improves safety, reduces risk of thermal runaway, enhances structural stability.	Enhances safety and stability, good rate capability, extends battery lifespan by mitigating capacity fading.	Relatively low specific energy and specific strength.	(Koech et al., 2024, Yang et al., 2023, Schipper et al., 2017, Satpathy and Pamuru, 2021, Xu et al., 2012)
Cobalt	Enhances thermal stability, improves low-temperature performance, increases energy density, and optimizes rate performance.	Improves thermal stability and low-temperature performance, increases energy density and rate performance.	High cost, environmental and ethical concerns related to mining.	(Chu et al., 2022, Cunningham, 2022)
Aluminium	Provides stability in NCA chemistries, contributes to long service life, maintains good specific power strength.	Increases structural stability and battery life, comparable specific energy to NMC, lightweight and cost-effective.	Typically used in small quantities (5 %-10 %), limited to specific applications.	(Satpathy and Pamuru, 2021, Lebens-Higgins et al., 2019, Julien and Mauger, 2020)

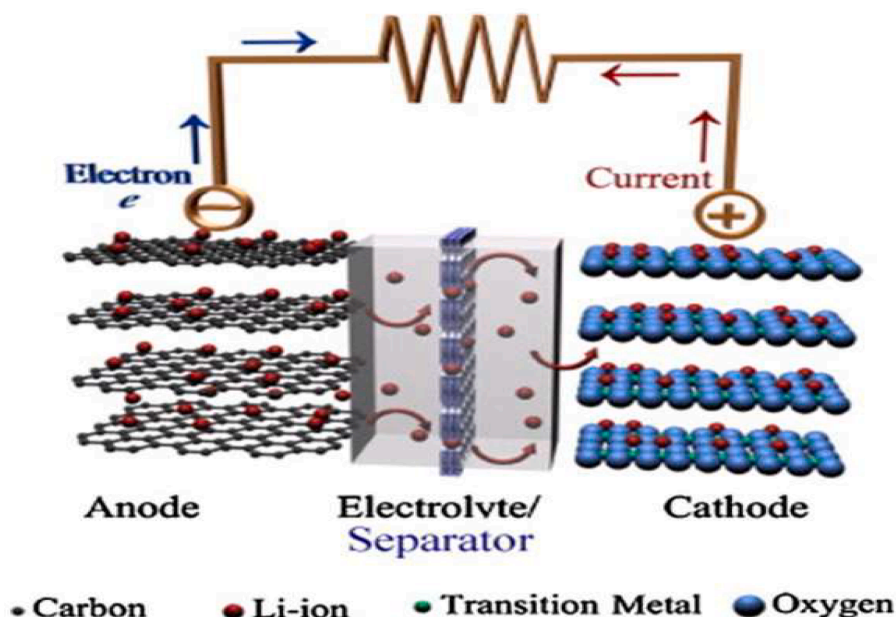


Fig. 1. Illustration of the basic components and operating principle of Li-ion batteries. (Adapted from Lyu et al., 2019).

limited by low Coulombic efficiency and significant hysteresis (Fang et al., 2020). Alloys, on the other hand, offer exceptional energy storage capabilities ($400\text{--}2300\text{ mA h g}^{-1}$), impressive durability over multiple cycles, and satisfactory Coulombic efficiency but suffer from reduced conductivity, substantial volume variation, and a capacity decrease (Hossain et al., 2023). Carbonaceous materials, particularly graphite, carbon, and graphene, are the most commonly used anode materials in commercial Li-ion batteries, delivering a capacity of 372 mA h g^{-1} due to the formation of LiC_6 (Ding et al., 2020). These carbon anodes are favored for their enhanced safety, excellent structural integrity, superior conductivity, and long lifespan (Nzereogu et al., 2022, Li et al., 2024). According to Ding et al. (2020), the high initial coulombic efficiency, low volume expansion, abundant natural availability, and ability to form graphite intercalation compounds make graphite a widely used anode material not only in Li-ion batteries but also in potassium-ion and other alkaline metal-ion batteries, such as rubidium and cesium ion batteries (Ding et al., 2020). Nevertheless, carbon anodes are constrained by their moderate capacity, occasional safety concerns, and reduced discharge rates (Nzereogu et al., 2022, Ding et al., 2020, Li et al., 2024).

3.2. Binder technologies in Li-ion batteries

The choice of binder is crucial for the effectiveness of anode materials in batteries due to its role in maintaining the mechanical integrity and electrochemical stability of the electrode. A well-selected binder ensures that the active material particles, conductive additives, and current collector remain cohesive, preventing disintegration during battery operation and preserving electrical contact (Liu et al., 2017). Moreover, the binder must be electrochemically stable to avoid degradation, which could lead to unwanted side reactions that reduce the battery's capacity and cycle life (Ajdari et al., 2024). Additionally, a good binder contributes to the formation of a stable solid electrolyte interphase (SEI) layer, enhancing cycling stability and overall durability. Its chemical compatibility with the electrode and electrolyte materials is essential to prevent issues like swelling or dissolution that could compromise performance and safety (Sun et al., 2024). The binder should be flexible enough to accommodate the volume expansion of certain anode materials, such as silicon, during lithiation, without causing cracks or loss of adhesion (Tareq and Rudra, 2024). Finally,

considerations like ease of processing, environmental impact, and the ability to maintain adhesion over many charge/discharge cycles are vital for the overall contribution of the binder to the durability and effectiveness of the battery (Yoon et al., 2024).

The typical binders utilized in anode applications include water-based polymers such as alginates (ALG), carboxymethyl cellulose (CMC), polyacrylic acid (PAA), styrene-butadiene rubber (SBR), and other water-soluble polymers. A commonly employed combination, even in industrial settings, involves CMC and SBR, appreciated for their eco-friendly attributes and cost-effectiveness (Shin et al., 2017, Jagau et al., 2022). For instance, while CMC serves as a binder solution to attain a stable slurry with desirable viscosities for coating due to its carboxylic groups, SBR functions as an emulsifier and shear-sensitive binder crucial for the mechanical stability of the electrode (Jagau et al., 2022). The anode typically constitutes approximately 15–30 % of the total battery weight, including a copper collector (Łukasz et al., 2023).

The active materials of the electrode are combined with high-surface-area carbon black to reduce electrical resistance and thereby enhance conductivity (Entwistle et al., 2022). Additionally, a polymeric binder, typically polyvinylidene fluoride (PVDF), constitutes a small portion of the electrode material (usually 2–5 % of the total mass in commercial electrodes), with N-methyl-pyrrolidone (NMP) acting as the solvent. NMP is selected for use because the PVDF binder dissolves only in NMP, not water. The binder is used in the cathode to maintain contact between all components and ensure the uniform distribution of active materials by aiding in the dispersion of other materials in the solvent during the fabrication process. Also, the binder can be utilized as a thickening agent (Casimir et al., 2016). Furthermore, the binder amalgamates all materials—including the active components and any conductive additives—onto the current collector, ensuring that every component is in contact and providing mechanical stability to the electrode (Chen et al., 2024). Fig. 2 presents a diagram illustrating a composite electrode with a binder interacting with its different components. The binder connects these elemental components through chemical or physical interactions, maintaining the mechanical integrity of the electrode without significantly affecting its electronic or ionic conductivity. Additionally, binders can serve as an interface between the electrode and electrolyte, shielding the electrode from corrosion or preventing electrolyte depletion while enabling ion transport across the

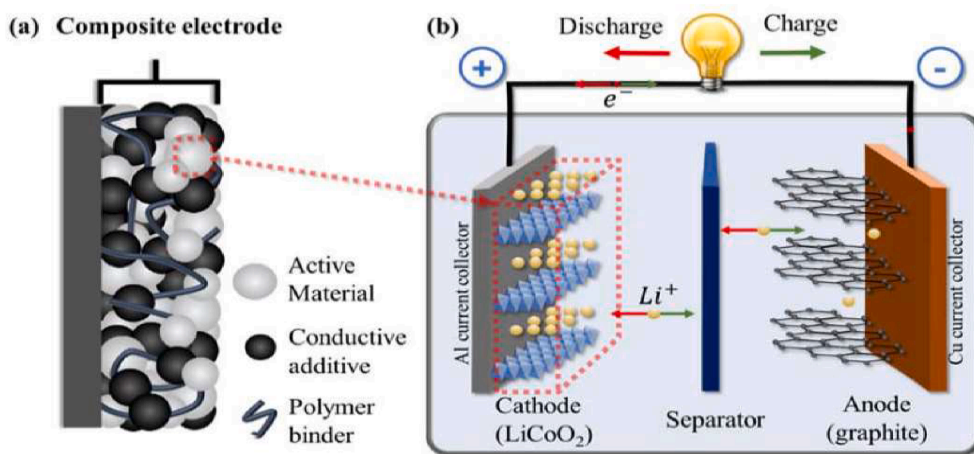


Fig. 2. Diagram (a) illustrates a composite electrode comprising the active material, conductive additive, and polymeric binder attached to a current collector. (b) A Li-ion battery with an LCO cathode and an anode made of graphite during discharge (the reactions taking place within a crystallite of active material being shown) (Cholewinski et al., 2021).

interface (Cholewinski et al., 2021, Yoon et al., 2024).

3.3. Electrolyte composition and additives in Li-ion batteries

The major source of positive lithium ions essential for battery operation is the dissolved lithium salts within the electrolyte. The movement of electrons between the negative and positive current collectors is facilitated by their migration to and from the anode and cathode via the electrolyte and separator (Whitehead and Schreiber, 2005). In terms of composition, lithium is present in the form of conductive salts (electrolytes) such as lithium hexafluorophosphate (LiPF_6), lithium bis(trifluoromethylsulfonyl)imide $\text{LiC}_2\text{F}_6\text{NO}_4\text{S}_2$, Lithium tetrafluoroborate (LiBF_4), Lithium triflate ($\text{Li}(\text{CF}_3\text{SO}_3)$), lithium perchlorate (LiClO_4), or lithium hexafluoroarsenate (LiAsF_6), which make up approximately 1.2 to 2.0 % of the overall mass of the battery (Jagau et al., 2022). These crystalline salts dissolve in low-viscosity, highly conductive solvents like linear carbonates such as diethyl carbonate (DEC), dimethyl carbonate (DMC), and ethyl methyl carbonate (EMC) or cyclic carbonates like ethylene carbonate (EC), propylene carbonate (PC), and γ -butyrolactone (γBL), particularly in the case of liquid electrolytes (Park et al., 2021, Anouti et al., 2013).

During the charging and discharging processes, the organic solvent in the electrolyte plays a crucial role in dissolving and transporting lithium ions between the anode and cathode. Due to the high reactivity of pure metals, non-aqueous electrolytes are commonly used in EV batteries to prevent adverse reactions, such as the vigorous production of hydrogen gas and lithium hydroxide (LiOH) when pure lithium contacts water (Koech et al., 2024). To ensure stability, the electrolyte is designed to be chemically inert with other cell components, including separators, electrode substrates, and packaging materials. Additionally, it must provide high ionic conductivity while remaining electronically insulating to enable efficient ion transfer without self-discharge and maintain electrochemical stability to avoid degradation within the battery's operating voltage range (Koech et al., 2024, Minkiewicz et al., 2023).

The selection of a liquid electrolyte involves considering factors such as low viscosity for easier ion transport, a high dipole moment for effective salt dissolution, thermal stability, and non-toxicity to ensure overall performance, reliability, and safety (Dhanda et al., 2022). These electrolytes often incorporate various additives, including cathode protection agents that prevent degradation (e.g., dimethyl acetamide, silicones, triethyl phosphite (TePi), tris(trimethylsilyl) phosphite (TMSPI), lithium difluoro(oxalate)borate (LiDFOB), and tributylamine) (Park et al., 2021, Anouti et al., 2013, Michan et al., 2016). Additionally,

there are enhancers for lithium deposition, such as fluoroethylene carbonate (FEC), stabilizers that protect against the decomposition of lithium salts, compounds for shutdown, like cyclohexylbenzene or biphenyl, overcharge protection substances such as anisole derivatives, surface-film-forming additives like vinylene or vinylene carbonate, and fire-retardants (Park et al., 2021, Anouti et al., 2013, Michan et al., 2016, Tornheim et al., 2017). The additives, such as vinylene carbonate, sulfone, and cyclic sulfate, also act as coordinators within the electrolyte, altering the interaction between lithium ions and solvents. Consequently, these additives dictate the behaviour of lithium ions or clusters, including competitive insertion or removal (intercalation/deintercalation) of lithium ions, as well as simultaneous insertion (co-intercalation) of lithium ions and solvents into the anode electrode (Ming et al., 2019). Additionally, these additives play a crucial role in preventing a decrease in solvent fluidity, especially in colder temperatures, which could otherwise lead to a reduction in the performance of the Li-ion cell. While solid inorganic technologies (e.g., lithium lanthanum zirconate ($\text{La}_3\text{Li}_7\text{O}_{12}\text{Zr}_2$), sodium superionic ($\text{Li}_{1.5}\text{Al}_{0.3}\text{Si}_{0.2}\text{Ti}_{1.7}\text{P}_{2.8}\text{O}_{12}$) type, amorphous lithium phosphorous oxynitride sulfidic glasses) or polymer electrolyte technologies (solvent-free polyethylene oxide), aided by a variety of additives, are gaining traction in the market, liquid electrolytes remain the most widely used (Na et al., 2023, Hu et al., 2022, Inoue et al., 2022).

3.4. Separator technologies in Li-ion batteries

The separator, typically a thin microporous polymer membrane, plays a crucial role in Li-ion batteries by facilitating ionic transport within the cell and acting as an electrolyte reservoir, isolating or preventing physical contact between the negative and positive electrodes (Pan et al., 2017). This isolation enables the exchange of lithium ions while preventing the flow of electrons, thus insulating against internal short-circuit occurrences (Heidari and Mahdavi, 2019, Zhang et al., 2014). It is predominantly fabricated from microporous materials such as polypropylene (PP), polyethylene (PE), or a combination of the two (PE/PP membrane) (Heidari and Mahdavi, 2019, Gu et al., 2021). These barrier materials, composed of semi-crystalline polyolefins or made from alternative materials such as non-woven fabric mats, inorganic composites, or their modified forms, are chosen for their optimal balance of permeability, mechanical strength, and electrochemical inertness, which are essential for effectively halting the migration of electrode particles and retaining a sufficient amount of electrolyte (Gu et al., 2021). Thus, these separators are wetted by the liquid electrolyte and placed between the two terminals (the cathode and anode) to

enhance the overall safety, performance, and cost-effectiveness of Li-ion batteries, owing to their high electrochemical reaction inertness, mechanical strength, and reasonable cost (Heidari and Mahdavi, 2019, Lee et al., 2014). To ensure optimal performance, the membrane should have large and interconnected pores inside to enable high ionic conductivity and a flat surface outside to prevent lithium dendritic growth as well as short-circuiting. This design addresses safety concerns and ensures a long and sufficient cycle life for Li-ion batteries (Rafiz et al., 2022). PE and PP polyolefin membranes have a lower melting point and greater dimensional shrinkage due to the melting of the pore walls, increasing the risk of internal electrical short circuits, fire, or even explosion (Zhang, 2007). Consequently, other polymers have also been utilized to fabricate microporous membranes, including isotactic poly (4-methyl-1-pentene), PE–PP blend, polyoxymethylene, polystyrene (PS)–PP blend, and poly(ethylene terephthalate) (PET)–PP blend (Hao et al., 2021). Despite recent advancements in solid electrolytes or gel polymer electrolytes, which can improve thermal stability to some extent, batteries using these electrolytes often exhibit low ionic conductivity or weak mechanical strength, thereby limiting their widespread use (Dhanda et al., 2022). Other separators considered for use are those proposed by Rafiz et al. (2022) (Li and Duan, 2023), who utilized electrode-coated plate-structured γ -alumina separators with liquid electrolytes to ensure safe and high-performing Li-ion batteries. These separators are made of γ -alumina plates with a high aspect ratio, making them mechanically robust and tortuously porous, effectively preventing the passage of dendrites (Li and Duan, 2023).

3.5. Key considerations in Li-ion battery design and environmental impact

Other additional materials in a battery include a casing made of either a Fe–Ni alloy, aluminium, or plastic (Guo et al., 2021). While the material used for the container does not impact the properties of the battery, it is composed of easily recyclable and stable compounds. The anode, cathode, separator, and electrolyte are crucial for the cycling process (charging and discharging) of the cell. However, if not handled properly, these components—particularly the metals in the cathode—can have notable environmental and health implications (Łukasz et al., 2023).

The battery assembly is done in a discharged state, with all lithium ions residing in the oxide host at the cathode. Lithium ions deintercalate and travel from the host cathode structure (transition metal oxide) to the anode, where they are inserted into the graphite when an electrochemical potential difference is applied (Guo et al., 2021, Borah et al., 2020). Ions from the cathode are elevated or shuttled to a higher energy state at the anode (Guo et al., 2021, Maleki and Howard, 2004). During discharge, lithium ions leave the graphite and re-intercalate into the metal oxide host structure, the cathode. Simultaneously, electrons are released into an external circuit to balance the charge. These intercalation reactions involve the transfer of lithium ions and the associated generation and consumption of electrons, which provide electrical energy for the device or system while maintaining charge balance within the battery (Guo et al., 2021). Table 2 below summarizes the key components of Li-ion batteries, along with their main features, challenges,

Table 2
Key components of Li-ion batteries: features, challenges, and future developments.

Component	Key materials/technologies	Main features	Challenges	Future developments	References
Cathode	LCO, LMO, NMC, NCA, LFP, High-voltage spinels, High-capacity layered Li-/Mn-rich oxides	Provides energy storage, contributes to capacity and voltage, influences battery stability and lifespan	Capacity fading due to Mn^{3+} instability, resource limitations, high cost, safety concerns	Doping strategies, advanced coatings, high-nickel NMC, cobalt-free/reduced materials, Mn-rich cathodes, zirconium doping	(Guo et al., 2021, Zhang et al., 2023, Chen et al., 2022, Gao et al., 2015, Muller et al., 2021)
Anode	Graphite, Silicon composites, Transition Metal Oxides (TMOs), Alloys, Lithium metal anodes	Stores lithium ions, affects battery capacity, influences charge/discharge rates	Limited capacity, potential for dendrite formation, safety concerns, volume expansion	Silicon anodes, hybrid anodes, Li-metal anodes, improved SEI formation, TMOs, and alloy-based anodes	(Guo et al., 2021, Hossain et al., 2023, Szczec and Jin, 2010, Guo et al., 2019, Liu et al., 2019, Fang et al., 2020, Ding et al., 2020)
Electrolyte	Liquid electrolytes (e.g., LiPF ₆ in organic solvents), Solid-state electrolytes, Additives	Facilitates ion transport between cathode and anode, impacts safety and performance	Flammability, thermal stability, ion conductivity, electrolyte instability	Solid-state electrolytes, gel electrolytes, non-flammable electrolytes, electrolyte additives for stability	(Koech et al., 2024, Whitehead and Schreiber, 2005, Park et al., 2021, Anouti et al., 2013, Minkiewicz et al., 2023, Dhanda et al., 2022, Michan et al., 2016, Tornheim et al., 2017, Ming et al., 2019, Na et al., 2023, Hu et al., 2022, Inoue et al., 2022), (Pan et al., 2017, Heidari and Mahdavi, 2019, Zhang et al., 2014, Gu et al., 2021, Lee et al., 2014, Rafiz et al., 2022, Zhang, 2007, Hao et al., 2021, Li and Duan, 2023), (Mosa and Aparicio, 2018, Lamiel et al., 2022, Khatibi et al., 2022),
Separator	Polyethylene (PE), Polypropylene (PP), Ceramic-coated separators, γ -Alumina separators	Prevents short circuits, maintains structural integrity, influences battery safety and longevity	Thermal stability, mechanical integrity, risk of internal short circuits due to melting or shrinking	Enhanced thermal stability, thinner separators, ceramic or composite separators, electrode-coated separators	(Pan et al., 2017, Heidari and Mahdavi, 2019, Zhang et al., 2014, Gu et al., 2021, Lee et al., 2014, Rafiz et al., 2022, Zhang, 2007, Hao et al., 2021, Li and Duan, 2023), (Mosa and Aparicio, 2018, Lamiel et al., 2022, Khatibi et al., 2022),
Current Collectors	Aluminium for cathode, Copper for anode	Conducts electricity between electrode and external circuit, distributes current uniformly	High cost of metals, corrosion, compatibility with electrode materials	Development of lightweight, corrosion-resistant materials, coatings for current collectors	(Liu et al., 2017, Ajdari et al., 2024, Tareq and Rudra, 2024, Yoon et al., 2024, Shin et al., 2017, Jagau et al., 2022, Casimir et al., 2016, Chen et al., 2024), (Ding et al., 2020)
Binder	Polyvinylidene fluoride (PVDF), Carboxymethyl cellulose (CMC), Styrene-butadiene rubber (SBR)	Maintains mechanical integrity of the electrode, ensures adhesion between materials	Electrochemical stability, flexibility to accommodate volume changes, processability	Development of flexible and durable binders, improved SEI formation, eco-friendly binders	(Liu et al., 2017, Ajdari et al., 2024, Tareq and Rudra, 2024, Yoon et al., 2024, Shin et al., 2017, Jagau et al., 2022, Casimir et al., 2016, Chen et al., 2024), (Ding et al., 2020)
Battery management system (bms)	Sensors, microcontrollers, software algorithms	Monitors and optimizes battery performance, ensures safety	Complexity, reliability, cost	Advanced BMS with AI integration, real-time monitoring	(Muller et al., 2021, Łukasz et al., 2023)
Battery material recycling strategies	Lithium and critical material recovery processes	Ensures sustainable supply chain, reduces environmental impact, contributes to resource conservation	Efficiency, scalability, cost	Enhanced recycling techniques, closed-loop processes, improved material recovery efficiency	(Muller et al., 2021, Łukasz et al., 2023)

and future developments.

4. Comparative analysis of Li-ion battery chemistries for EVs

Li-ion batteries have become the cornerstone of EV technology due to their superior specific energy, longevity, and energy density, making them ideal for providing the necessary range and performance (Miao et al., 2019). Fig. 3 illustrates a comparison of various Li-ion battery types used in EVs, evaluating several critical characteristics (Wang et al., 2016). These Li-ion battery compositions—such as LFP, LCO, LMO, LTO, NMC, and NCA—each offer distinct advantages and trade-offs, making them suitable for different applications. A larger coloured area in the performance charts indicates a more favourable performance profile, encompassing factors like cost, specific energy, specific power, lifespan, and overall performance (Miao et al., 2019).

NMC batteries are popular in EVs due to their well-rounded performance across multiple metrics, offering a good balance of energy density, power, and lifespan. In contrast, LFP batteries, despite having lower energy density, are favoured for their superior safety, thermal stability, and extended cycle life (Zhang et al., 2023). NCA batteries, known for their high energy density and specific power, face challenges such as higher costs, shorter lifespans, and the need for robust thermal management, which can impact safety (Itani and Bernardinis, 2023). This comparison underscores the importance of selecting a battery chemistry based on the specific requirements of the application, balancing performance, cost, and safety considerations.

Among the six leading Li-ion battery chemistries, NMC, LFP, and Lithium Manganese Oxide (LMO) are recognized as superior candidates. These materials excel due to their unique balance of energy density, power density, safety, and overall performance, making them particularly suitable for applications like EVs and energy storage systems (Fig. 3) (Miao et al., 2019).

5. Energy density and enhancements

For electric vehicles (EVs), driving range is a key concern for customers, largely determined by the battery's energy density—the amount

of energy stored per unit volume or weight (Deng et al., 2020). According to Nadim Maluf (2020), the driving range of an EV is directly proportional to its total battery capacity (Maluf, 2020). Khan et al. (2023) emphasize that Li-ion batteries have significantly improved EV performance due to their high energy storage density and efficiency (Nizam Uddin Khan et al., 2023). To address the growing demand for longer driving ranges on a single charge, there is an urgent need for advancements in batteries with higher energy densities, enabling EVs to travel greater distances per charge (Olabi et al., 2022). Advancements may also include technologies such as solid-state batteries, lithium-sulfur batteries, lithium-air batteries, and magnesium-ion batteries. Such innovations hold the potential to extend the range and enhance the performance of EVs while reducing the frequency of recharging (Deng et al., 2020, Nizam Uddin Khan et al., 2023).

Currently, Li-ion batteries exhibit some of the highest energy densities, ranging from 250 to 693 Wh L⁻¹ (100 to 265 Wh kg⁻¹), and power densities of up to 340 W kg⁻¹, with a lifespan exceeding 1,000 cycles (El Kharbachi et al., 2020, Daniel, 2015). However, despite these impressive figures, their energy density is still significantly lower than that of gasoline. Consequently, multiple batteries are needed to achieve a driving range of 200–300 miles (322–483 kilometers), which is comparable to gasoline-powered vehicles (Nizam Uddin Khan et al., 2023). To bridge this gap, researchers are developing next-generation batteries with even higher energy densities than Li-ion batteries, aiming for energy densities greater than 750 Wh L⁻¹ (350 Wh kg⁻¹). Among these, NMCs (Nickel Manganese Cobalt) have been identified as promising candidates, with a potential energy density of approximately 900 Wh kg⁻¹. Additionally, conversion-type alternative cathode materials, such as FeF₂ and FeF₃, theoretically offer ultrahigh energy densities of up to 1500 Wh kg⁻¹, though they have not yet been realized in practice due to significant capacity fading (Nizam Uddin Khan et al., 2023). Advancements may also include technologies such as solid-state batteries, lithium-sulfur batteries, lithium-air batteries, and magnesium-ion batteries. Such innovations hold the potential to extend the range and enhance the performance of EVs while reducing the frequency of recharging (Deng et al., 2020, Nizam Uddin Khan et al., 2023).

Emerging technologies in battery development offer several

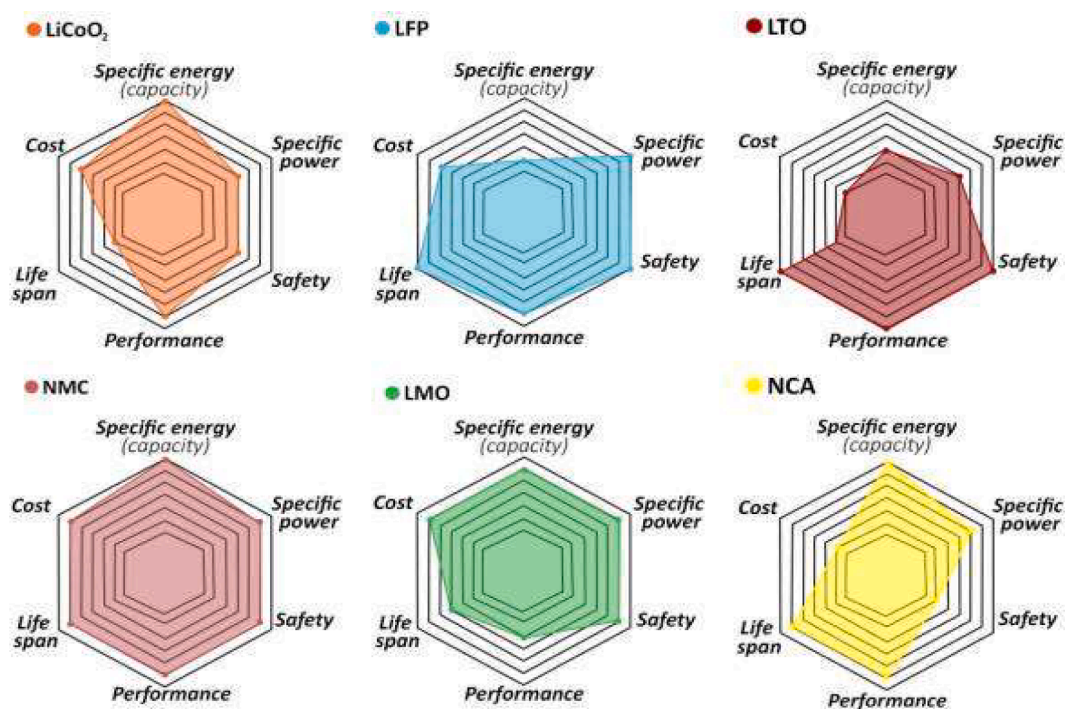


Fig. 3. Comparisons of different types of Li-ion batteries used in EVs from the following perspectives: specific energy (capacity), specific power, safety, performance, life span, and cost (the outer hexagon is most desirable) (Miao et al., 2019, Salgado et al., 2021).

promising advancements: i) Solid-state batteries, utilizing a solid electrolyte instead of a liquid or gel, promise higher energy densities ranging from 0.3 to 0.5 kWh kg⁻¹, improved safety, and a longer lifespan due to reduced risk of dendrite formation and thermal runaway (Moradi et al., 2023); ii) Lithium-sulfur batteries, with potential energy densities reaching up to 0.6 kWh kg⁻¹, offer significant improvements over current Li-ion batteries but face challenges like the polysulfide shuttle effect and capacity fading (Zhang et al., 2018); iii) Lithium-air batteries, theoretically providing the highest energy densities among all battery types, range from 1.7 to 2.0 kWh kg⁻¹, though challenges related to oxygen reactivity and electrolyte stability must be addressed (Chen et al., 2022); and iv) Magnesium-ion batteries, exploring a safer and more energy-dense alternative to Li-ion batteries, have potential energy densities around 0.4–0.5 kWh kg⁻¹, but face hurdles in developing suitable electrolytes and cathode materials for efficient magnesium-ion conduction (Mao et al., 2019).

Despite these advancements, Li-ion batteries remain the dominant technology in the energy storage industry, widely used in handheld and portable electronics as well as EVs. Their popularity is due to their high specific energy (Wh kg⁻¹), long cycle life, and superior efficiency (Gao et al., 2022, Yuan et al., 2021, Quarti et al., 2022). Fig. 4 illustrates the significant advantages of Li-ion batteries over other types, including nickel-metal hydride (Ni-MH), nickel-cadmium (Ni-Cd), lead-acid, lithium-ion polymer (Li-Po), and lithium metal batteries (Gao et al., 2022, Mahmud et al., 2022). As the volumetric energy density increases from 0 to 600 Wh L⁻¹ along the X-axis, the size of the battery material decreases, while on the Y-axis, the gravimetric energy density (Wh kg⁻¹) increases, resulting in lighter materials. This trend indicates that Li-polymer and Li-ion batteries not only become smaller in size but also lighter (Zhou, 2016, Daud et al., 2013). However, despite these advantages, Li-ion batteries are not without their drawbacks, including high costs and the necessity for complex safety and monitoring systems (Miao et al., 2019).

4. Cathode materials

The positive electrode, known as the cathode, in a cell is associated with reductive chemical reactions. This cathode material serves as the primary and active source of most of the lithium ions in Li-ion battery chemistries (Tetteh, 2023). The preferred choice of positive electrode materials, influenced by factors such as performance, cost, and safety considerations, depends on whether it is for rechargeable lithium-metal or Li-ion batteries (Fig. 5) (Tarascon and Armand, 2001, Jiang et al., 2022). Fig. 5 provides an overview of Li-ion battery materials,

comparing the potential capabilities of various anode and cathode materials. Among these, lithium exhibits the highest specific capacity; however, its use is limited due to the increased risk of cell explosiveness and dendrite formation (Kurc et al., 2021). The lithiation/delithiation potentials of different materials are shown relative to the Li/Li⁺ redox potential, which is set at 0V. The specific capacity of these materials, representing their ability to store charge in the form of lithium ions, is measured in A h kg⁻¹ (equivalent to 3.6 C g⁻¹) (Brumbarov, 2021). Since lithium metal functions as a negative electrode in rechargeable lithium-metal batteries, lithiation of the positive electrode is not necessary. In Li-ion batteries, however, since the carbon electrode acting as the negative terminal does not contain lithium, the positive terminal must serve as the source of lithium; hence, an intercalation compound is necessary during cell assembly (Tarascon and Armand, 2001).

Lithium layered cathode materials, such as LCO, LMO, LFP, NCA, and NMC, find application in Li-ion batteries. Among these, LCO, LMO, and LFP are the most widely employed cathode materials, along with various other lithium-layered metal oxides (Heidari and Mahdavi, 2019, Zhang et al., 2014). Each of these cathode materials provides different levels of energy density, thermal stability, and cost-effectiveness (Pillai et al., 2023, Saa'id et al., 2024). When evaluating the cost, durability, efficiency, safety, energy density, and power of these materials, NMC emerges as the top performer in many aspects. Accordingly, due to its numerous advantages, the NMC battery is becoming increasingly popular in the global Li-ion battery market as researchers continue to advance its design characteristics (Saa'id et al., 2024). These cathode materials, based on lithium intercalation, have the capability to efficiently and reversibly accept and eject lithium ions into and out of their crystal structure throughout charge/discharge cycles (Ni et al., 2014). For stationary applications, where weight is less of a concern, nickel-metal hydride (Ni-MH) technologies may be considered due to their ability to withstand deep discharge conditions. These Ni-MH batteries commonly utilize a cathode material composed of nickel oxyhydroxide (NiOOH). During the charging process, the cathode absorbs hydroxide ions and subsequently releases them during the discharge process (Ash et al., 2020). A separator, typically made from synthetic nonwoven material, is used as an insulator between the anode and cathode in Ni-MH batteries. It also functions as a medium for absorbing electrolytes (Fan et al., 2020). The battery chemistry of Ni-MH batteries is illustrated in Fig. 6.

On the other hand, lithium-air batteries utilize a porous, carbon-based cathode to enable interaction with oxygen from the surrounding air. The battery functions through the catalytic reduction of oxygen in an alkaline aqueous electrolyte and metallic lithium in a non-aqueous

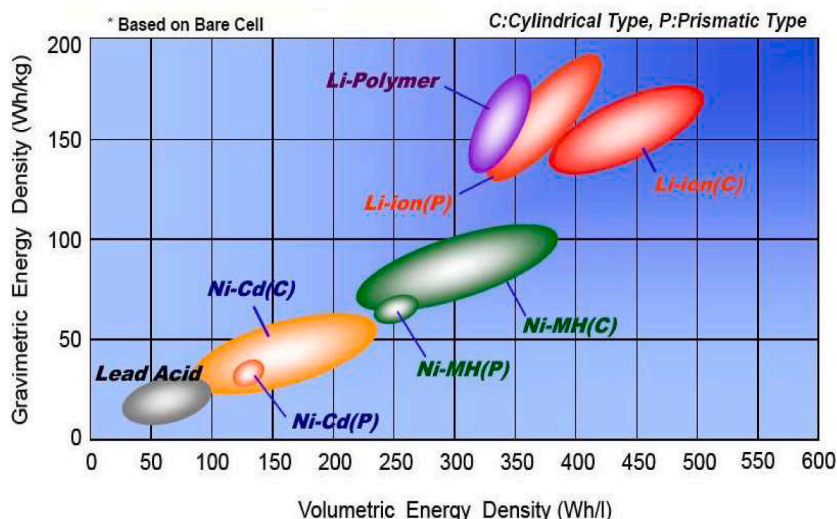


Fig. 4. Ragone plot showing comparison of the energy density of several of the battery technologies at cell level used in EVs (Beggi et al., 2018).

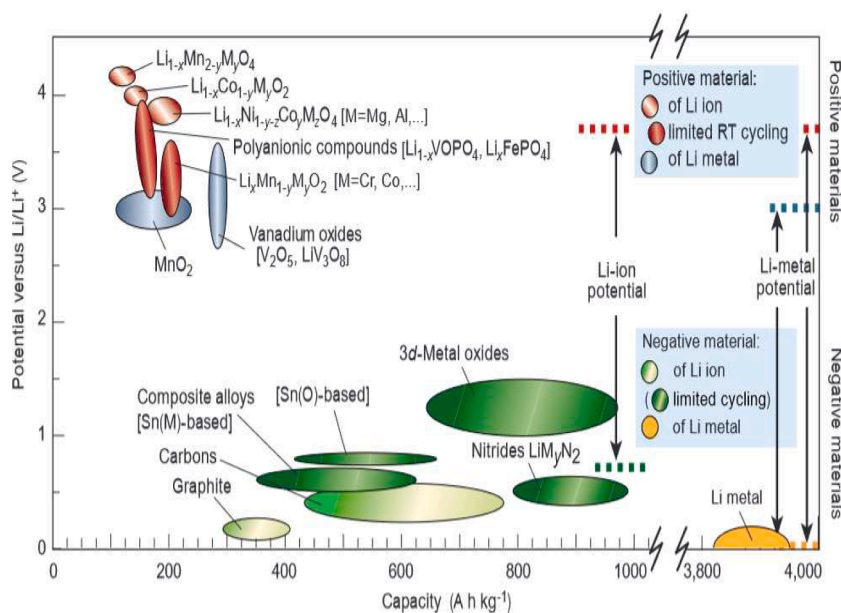


Fig. 5. Illustrates the voltage (V) versus capacity (A h kg^{-1}) for current and potential future positive- and negative-electrode materials in rechargeable lithium-assembled cells. The graph displays output voltage values for both Li-ion and lithium metal cells. Notably, a significant capacity disparity exists between lithium metal and other negative electrodes, highlighting lithium metal as the best potential option and driving continued interest in resolving dendrite growth issues (Tarascon and Armand, 2001).

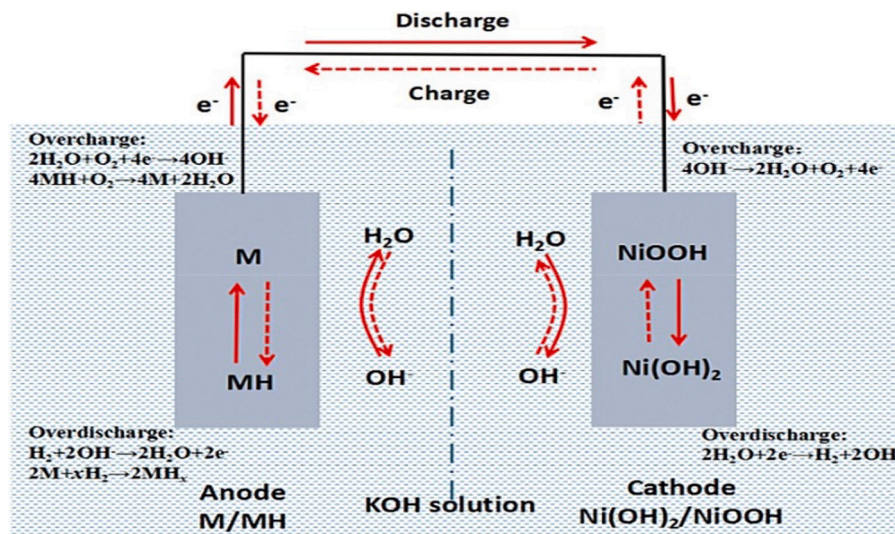


Fig. 6. Schematic of Ni-MH batteries (Adapted from Fan et al., 2020).

electrolyte, such as a solid ceramic polymer electrolyte, glass, or glass-ceramic electrolyte (Wang and Zhou, 2010, Capsoni et al., 2015, Imanishi and Yamamoto, 2019). The aqueous lithium-air cell configuration is illustrated in Fig. 7 (Imanishi and Yamamoto, 2019).

This configuration enables a reversible electrochemical reaction between Li-ions and oxygen to occur seamlessly throughout the charge and discharge cycles. It's important to note that the cathodic reaction in lithium-air batteries is electrocatalytic in nature, necessitating the presence of an electrocatalyst for optimal performance (Wang and Zhou, 2010).

4.1. Lithium manganese (Li-Mn-O) spinels

The discovery of Li-Mn-O, a compound with a research history spanning over 50 years for battery applications, stemmed from the quest

for a cost-effective oxide suitable for use as a cathode material. Further research into these complex phase diagrams, the versatile structure, and chemistry of manganese-based materials, alongside increased efforts to improve and optimize the electrochemical properties, particularly concerning cycling, has led to the identification and enhancement of high-voltage spinel cathodes such as $\text{LiNi}_{1/2}\text{Mn}_{3/2}\text{O}_4$, high-capacity layered Li-/Mn-rich cathodes like Li_2MnO_3 and $n\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2 \cdot (1-n)\text{Li}_2\text{MnO}_3$, as well as various other innovative cathode materials and composites (Huang et al., 2020). Chitrakar et al. (2001) suggest that the chemical delithiation of LiMn_2O_4 spinel in aqueous acidic solutions, leading to the production of $\lambda\text{-MnO}_2$, a distinct form of manganese dioxide, was documented in 1981. $\lambda\text{-MnO}_2$ retains the $[\text{B}_2]\text{O}_4$ framework of $\text{A}[\text{B}_2]\text{O}_4$ spinel and emerges as the final product of LiMn_2O_4 following electrochemical delithiation (Huang et al., 2020). Its various forms make it well-suited for accommodating small helium and lithium ions

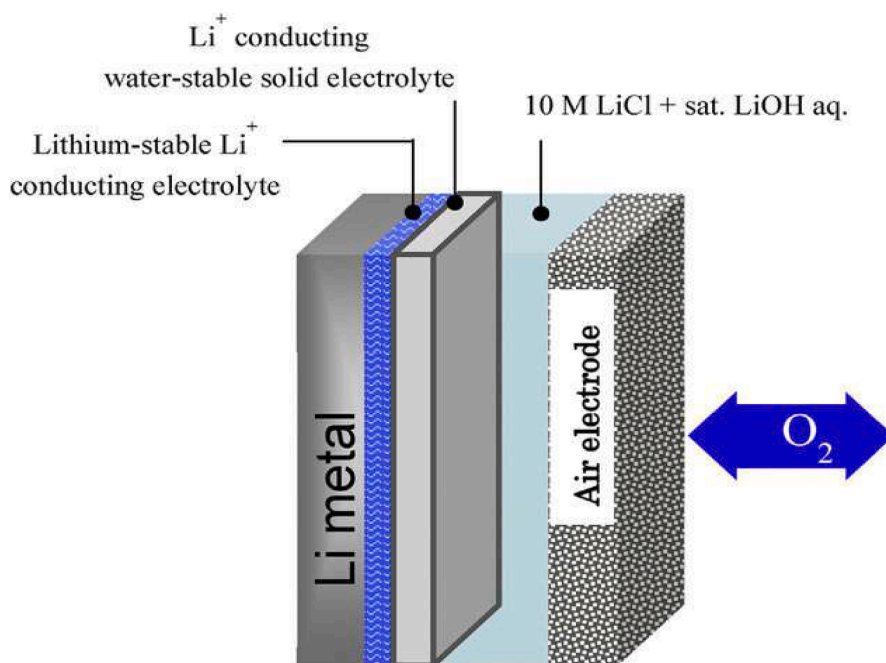


Fig. 7. Structure of the aqueous lithium-air battery. (Adapted from Imanishi and Yamamoto, 2019).

through intercalation. The intercalation of Li-ions is facilitated by the lambda form featuring its spinel structure (Mn_2O_4) (Zhang et al., 2000).

Initially employed as a depolarizer, Li-Mn-O exhibits a range of advantageous traits: ready availability, non-toxicity, cost-effectiveness, and desirable electrochemical properties. Moreover, Li-Mn spinels with three-dimensional (3D) Li-ion pathways offer benefits such as improved ionic conductivity, high thermal tolerance, good rate capability, and a flat voltage profile around 4 V, making them safer (with minimal detrimental impact on the environment and human health) and suitable for high-discharge applications such as power tools and hybrid vehicles (Huang et al., 2020, Tomon et al., 2022). In contrast to the expensive and toxic lithium-cobalt-based (Li-Co-O) and the more difficult-to-produce lithium-nickel-based (Li-Ni-O) alternatives both exhibiting lithium diffusion coefficients ranging from 10^{-8} to 10^{-14} cm^2/s (Liu et al., 2018, Thackeray et al., 2012, Xu et al., 2012, Rao et al., 2022, Xia and Lu, 2007, Rahim et al., 2022), lithium manganese (Li-Mn) stands out as the most popular and extensively used battery material (Salgado et al., 2021). The diffusion coefficients reported in the literature for LiMn_2O_4 electrodes obtained through various methods range from 10^{-6} to 10^{-14} sq. cm s^{-1} (Tang et al., 2008). However, challenges arise due to decreased capacity (capacity fading) with frequent cycling, caused by the instability of the electrochemically active Mn^{3+} ion, which is manifested by manganese dissolution, resulting from a combination of structural transformation and transition metal dissolution at higher temperatures (exceeding 55°C) (Banerjee et al., 2017, Ali et al., 2024). For instance, an undesired cubic-tetragonal phase transition promotes Jahn-Teller distortion in the spinel structure, leading to significant capacity loss during cycling below 3 V (Huang et al., 2021). This Jahn-Teller distortion, occurring during the discharge state, leads to a structural transformation from cubic LiMn_2O_4 to tetragonal $\text{Li}_2\text{Mn}_2\text{O}_4$, causing significant anisotropic volume changes (16 %) and resulting in structural damage (Tomon et al., 2022). In such scenarios, enhancements can be achieved by doping LiMn_2O_4 with specific metal ions (such as Al, Co, Cr, Fe, Mg, and Ni) and applying acid-resistant coatings to improve structural stability (Callegari et al., 2022). To mitigate the Jahn-Teller effect, Li et al. (2024) proposed an innovative heterostructure design that combines layered and spinel domains, effectively disrupting long-range Jahn-Teller ordering. Their research focuses on the development of a Fe-doped $\text{Na}_{0.55}\text{Mn}_2\text{O}_4 \cdot x\text{H}_2\text{O}$ cathode for zinc-ion

batteries, demonstrating how this design stabilizes the interlayer structure, alleviates the Jahn-Teller distortion, and improves the electrochemical stability of the material. This approach enhances both structural integrity and electrochemical cycling stability (Li et al., 2024). Callegari et al. (2022) propose an approach to enhance the performance of LiMn_2O_4 cathodes by employing a dual-doping strategy. They suggest substituting Mn^{3+} with Fe^{3+} and Ti^{4+} to fabricate new active materials, termed $\text{LiFe}_x\text{Mn}_{2-x-y}\text{Ti}_y\text{O}_4$, with a Li-/Mn ratio varying between 1 and 1.7. This substitution effectively mitigates Jahn-Teller distortion, which is particularly noticeable in Ti-doped LiMn_2O_4 , allowing for cycling across a broader voltage range of 4.8 to 1.5 V and consequently resulting in increased capacity and enhanced stability (Callegari et al., 2022).

4.2. Lithium metal oxides

The focus on advancing NMC battery materials for electric vehicle (EV) applications is deliberate, given their exceptional energy density and operational voltage. These materials strike a balance between energy storage capacity, affordability, long driving range, and durability, while also providing the necessary high current during acceleration (Zeng et al., 2019, Roy et al., 2022). Lithiated nickel and cobalt oxides are among the most extensively researched cathode materials for Li-ion batteries due to their notable structural stability. However, their synthesis and production are challenging, primarily due to resource limitations, making them expensive. To address this, researchers have concentrated on developing solid materials with diverse compositions of active materials for the cathode within these layered compounds.

Different formulations of cathode materials, each featuring a unique combination of metals, offer distinct characteristics that influence their performance and suitability for various applications. For instance, NMC ternary battery materials, characterized by the general formula $\text{LiNi}_x\text{Mn}_y\text{Co}_{1-x-y}\text{O}_2$, represent a class of layered mixed metal oxides containing lithium, nickel, manganese, and cobalt. These materials are widely used in mobile devices, electronics, and EVs (Beggi et al., 2018, Malik et al., 2022). On the other hand, NCA comprises lithium-nickel-cobalt-aluminum oxide materials at its core, while LCO primarily contains lithium and cobalt (Beggi et al., 2018).

As one of the earliest commercial cathode materials, LCO is known

for its high capacity and fast charging capabilities, thanks to its excellent electronic conductivity, making it ideal for portable electronics like smartphones, laptops, and cameras (Chen et al., 2024). Despite its layered structure and high compact density of around 4.2 g cm^{-3} , LCO achieves only about half of its theoretical capacity of 137 mA h g^{-1} , primarily due to its relatively low operating voltage of 4.2 V (vs. Li^+/Li) under commercial conditions (Wang et al., 2021). Most improved LCO batteries typically operate at a nominal voltage of 3.7 V with a specific capacity of around $140\text{--}150 \text{ mA h g}^{-1}$ and moderate discharge rates, often between 0.2C and 1C (Chandra et al., 2023). The high energy density of LCO comes with trade-offs in safety and cycle life, as its thermal instability and susceptibility to thermal runaway—particularly under high temperatures or overcharging conditions—limit its lifespan to around $500\text{--}1000$ cycles (Hu et al., 2024). Additionally, the cobalt in LCO batteries degrades over time, leading to capacity loss. Despite these challenges, LCO batteries have become a staple in the consumer electronics industry due to their high energy density and performance. Nevertheless, their use in larger-scale applications, such as EVs, is limited by safety concerns and the high cost of cobalt (Krieger et al., 2013).

In contrast, lithium nickel cobalt aluminum oxide (LiNiCoAlO_2), commonly known as NCA, has been in use since 1999 for a wide range of applications, including EVs, energy storage systems, and consumer electronics. NCA is renowned for its long service life and high specific energy, comparable to NMC batteries. While NCA batteries excel in delivering good specific power, they face challenges in safety and cost, which remain obstacles to widespread adoption (Kurzweil and Garche, 2017). NCA's thermal sensitivity and reliance on cobalt, which raises cost and ethical concerns, present significant challenges. To address these issues, recent advances in NCA technology have focused on high-nickel variants with reduced cobalt content, along with coatings, doping, and nanostructuring techniques aimed at enhancing thermal stability, performance, and cycle life (Zhang et al., 2024). For instance, doping with elements like magnesium, titanium, or zirconium has been employed to suppress unwanted phase transitions and improve structural integrity under high-voltage conditions (Zhang et al., 2024). One example is the Zr-doped $\text{LiNi}_{0.98}\text{Co}_{0.07}\text{Al}_{0.03}\text{O}_2$ (Zr-NCA) cathode material, developed to address the rapid capacity decay of NCA at high working voltages. Through a nano-milling-assisted solid-state reaction, Zr-NCA demonstrates enhanced performance at 4.5 V , with an initial capacity of $225.9 \text{ mA h g}^{-1}$ and a coulombic efficiency of 89.29% . The strong Zr-O bonds suppress the irreversible H2-H3 phase transition, leading to a 22.84% improvement in capacity retention after 100 cycles at 0.5 C , and a promoted rate capability of $144.7 \text{ mA h g}^{-1}$ at 5 C . This doping approach shows promise for mass-producing high-energy-density cathode materials for EVs (Che et al., 2021). Similarly, NCA, with a high specific capacity of $180\text{--}200 \text{ mA h g}^{-1}$ and a layered structure was stabilized by aluminium doping, and offered superior energy density and a long cycle life exceeding 2000 cycles (Zhang et al., 2024). Surface coatings, such as Al_2O_3 and dual-layer $\text{LiAlO}_2/\text{Al}_2\text{O}_3$, have been applied to stabilize particle surfaces, protect against electrolyte degradation, and enhance Li-ion diffusion, resulting in improved cycle stability and capacity retention (Oh et al., 2004). In a study by Shana et al. (2020), an Al_2O_3 -modified coating on $\text{LiNi}_{0.98}\text{Co}_{0.09}\text{Al}_{0.03}\text{O}_2$ (NCA) particles was developed. This coating, initially believed to be pure Al_2O_3 , was found to have a dual-layer structure with LiAlO_2 forming in the inner layer and Al_2O_3 in the outer layer. The $\text{LiAlO}_2/\text{Al}_2\text{O}_3$ dual-modified coating enhances cycle stability and capacity retention, delivering a first charge/discharge capacity of $242.1/210.3 \text{ mA h g}^{-1}$ at 0.1 C , with improved performance even at higher current densities, significantly boosting the material's cycle life and rate performance (Shana et al., 2020).

Moving to NMC materials, the design of NMC electrodes prioritizes achieving high specific energy. The key to NMC's enhanced specific energy lies in the synergy between nickel and manganese. Nickel, known for its high specific energy, is counterbalanced by its poor stability,

while manganese, despite having low specific energy, is capable of forming spinel structures that contribute to low internal resistance. When combined with cobalt, these three transition metals exhibit a synergistic ternary effect, enhancing the overall performance of NMC. The specific blend of nickel, manganese, and cobalt varies among manufacturers and is often treated as proprietary information (Jung et al., 2019).

Layered Lithium Nickel-Manganese-Cobalt Oxide ($\text{LiNi}_x\text{Mn}_y\text{Co}_z\text{O}_2$ where $x + y + z = 1$) is a commonly utilized type of cathode material, with $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ (NMC 111 or NMC 333) being the most common basis composition, typically containing equal parts of nickel, manganese, and cobalt, each at 33% (Beggi et al., 2018). These compounds can be considered as solid solutions of LiCoO_2 and $\text{LiNi}_{0.5}\text{Mn}_{0.5}\text{O}_2$. In the latter, nickel is in the divalent form, and manganese is in the tetravalent form (Tian et al., 2017). In early studies, Shaju et al. (2002) reported layered $\text{Li}(\text{Ni}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3})\text{O}_2$, demonstrating a capacity of 160 mA h g^{-1} within the voltage range of 2.5 to 4.4 V . Yabuuchi and Ohzuku (2003) reported a capacity of approximately 200 mA h g^{-1} , with rate capability as high as 1.6 A g^{-1} within the voltage range of 2.5 to 4.6 V (0.17 mA cm^{-2}). NMC 111 possesses several favourable traits: it is cost-effective compared to LiCoO_2 , has a high theoretical capacity, low toxicity, good thermal stability, and a high energy density (Yoshizawa and Ohzuku, 2007). However, its extensive usage, especially in larger EV applications, is impeded by issues such as cation mixing and metal dissolution in the electrolyte during cycling (Lee et al., 2021). To address these hurdles, researchers are devoting extensive efforts to enhancing the electrochemical performance of NMC 111 cathode materials. These efforts include surface modification, ion doping, optimization of synthesis techniques, and the formulation of better composition designs (Zeng et al., 2018; Liao et al., 2022).

Moreover, current research is increasingly focusing on Nickel-rich NMCs (such as NMC 424, 532, 622, 712, 811, etc.) due to their superior specific capacity, energy characteristics (such as high energy density and long cycle life), and stability within the operational range (Wood et al., 2020; Gourley et al., 2020). Researchers are presently prioritizing the reduction of cobalt content in NMC, driven by ethical concerns surrounding cobalt mining and its high cost, scarcity, and centrally controlled and unpredictable supply, particularly highlighted in its use as a cathode raw material (Gent et al., 2022). The transition from NMC 111, which has a discharge capacity of 154 Ah kg^{-1} at 0.1 C , to these higher NMCs (nickel-rich NMC cathodes characterized by high mass-specific capacities, high-rate capabilities, and long-term cyclabilities) is motivated by the goals of enhancing battery discharge capacity, reducing reliance on cobalt, and achieving higher practical, specific capacities. This is particularly evident as specific capacities increase with an increase in nickel content (Li et al., 2019). NMC 811 ($\text{LiNi}_{0.8}\text{Mn}_{0.1}\text{Co}_{0.1}\text{O}_2$) has garnered significant attention due to its ability to deliver a higher discharge capacity ($>185 \text{ Ah kg}^{-1}$ at 0.1 C) and an average discharge potential of around 3.8 V (vs Li^+/Li), while also containing lower cobalt content (Miao et al., 2019). Furthermore, in general, practical specific capacities tend to increase with higher nickel content. For instance, NMC 811 exhibits specific capacities of around $190\text{--}200 \text{ mA h g}^{-1}$ compared to 160 mA h g^{-1} for NMC 442 when an upper cutoff voltage of 4.3 V is used in lithium half-cells (Tian et al., 2017). This makes NMC 811 a promising candidate as a positive electrode material for Li-ion batteries with high energy density (Zhang et al., 2018). A nickel-rich, low-cobalt NMC (with nickel content exceeding 90%) layered cathode is regarded as the optimal material due to its outstanding discharge capacity (exceeding 210 mA h g^{-1}) and energy density (over 740 Wh kg^{-1}), as well as its structural stability and other favourable properties (Luo et al., 2022). Recent studies by Nguyen et al. (2021) and Tian et al. (2023) have also highlighted the high-rate capability and excellent cycling stability of such cathode materials, making them promising candidates for next-generation Li-ion batteries.

However, adding more nickel beyond the optimal amount required in NMC can lead to issues such as high-capacity fading, as well as

inadequate structural and thermal stability, resulting in rapid cell failure (Ahangari et al., 2024). These degradation mechanisms arise from various factors, including (a) mixing of Ni^{2+} with Li^+ sites due to their similar ionic radii, hindering lithium diffusion; (b) undesirable side reactions with the electrolyte and ambient air, resulting in depletion of both the electrode and electrolyte; and (c) volume changes during cycling leading to the creation of microcracks. Nonetheless, these undesirable factors can be mitigated through methods such as thin surface coatings and ion doping. Surface coating primarily offers external protection for the materials, whereas ion doping alters their internal structure (Beggi et al., 2018, Jiang et al., 2021).

On the other hand, some researchers are opting to eliminate cobalt from these NMC batteries entirely (Noerochim et al., 2021). This is primarily due to concerns about cobalt's high cost, scarcity, and the centralized and unpredictable nature of its supply chain. However, compositions appropriate for EV applications, which exhibit both high energy density and long cycle life, rely on varying extents of cobalt or even its exclusion altogether. These compositions are currently known as nickel-rich cathode materials (Beggi et al., 2018, Gourley et al., 2020).

Muralidharan et al. (2020) reported on a new category of nickel-rich and cobalt-free layered cathode material characterized by the general formula $\text{LiNi}_x\text{Fe}_y\text{Al}_{1-x-y}\text{O}_2$ (NFA). They employed a co-precipitation reaction, successfully synthesizing $\text{NFA}(\text{OH})_2$ precursors wherein the constituent elements Ni, Fe, and Al were incorporated. According to their findings, the electrochemical performance assessments demonstrated that their cobalt-free material exhibited a high capacity of 190 mA h g^{-1} at 0.1C. Additionally, evaluations of cycling performance and rate capability indicated positive results, with capacity retention of 88 % at 100 cycles at C/3. By utilizing NFA cathodes, they also assembled a 0.5 Ah at C/3 cobalt-free Li-ion battery, which exhibited moderate cycling stability and, after 200 cycles, retained approximately 72 % of its capacity (Muralidharan et al., 2020). Lyu et al. (2015) examined $\text{Li}_{1.2}\text{Cr}_{0.4}\text{Mn}_{0.4}\text{O}_2$ (LiCrMO_2), an interesting intercalation-type cathode material. This material demonstrated a high theoretical capacity of 387 mA h g^{-1} due to multiple-electron transfer involving $\text{Cr}^{3+}/\text{Cr}^{6+}$, thus showcasing the high reversible capacity and structural stability achievable with LiCrMO_2 (Lyu et al., 2015). Wang et al. (2018), on the other hand, fabricated and studied a cation-disordered lithium-excess cathode material based on the binary system $x\text{Li}_3\text{NbO}_4-(1-x)\text{LiCrO}_2$, where $x = 0.43$, which was prepared successfully by mechanical milling. The highly reversible $\text{Cr}^{3+}/\text{Cr}^{6+}$ three-electron redox process involving the electrochemically inert niobium ion allowed them to attain a high reversible capacity of approximately 362 mA h g^{-1} (Wang et al., 2018).

Lithium Nickel Cobalt Oxide (LNCO), a two-dimensional positive electrode, is being considered for use in the newest generation of Li-ion batteries. Accordingly, LNCO exhibits remarkable thermal stability, along with high cell voltage and good reversible intercalation characteristics. It is typically readily available in varying volumes and quantities. The LNCO cathode fabricated by Sundaresan et al. (2007) exhibits a discharge capacity of around 100 mA h g^{-1} and retains approximately 63 % of its capacity over time. They introduced molybdenum doping, which substitutes certain cobalt ions in the LNCO structure, thus enhancing the discharging capacity to about 157 mA h g^{-1} and improving its ability to retain capacity (Sundaresan et al., 2007). Karupiah et al. (2020) investigated Layered $\text{LiNi}_{0.94}\text{Co}_{0.06}\text{O}_2$ (LNCO) as a potential energy storage material for both lithium-ion and sodium-ion (Na-ion) batteries, as well as for supercapacitor applications. Their analysis of the LNCO sample revealed favourable thermal stability, phase purity within the crystal structure, a notable Brunauer-Emmett-Teller (BET) surface area of $5.53 \text{ m}^2 \text{ g}^{-1}$, and a cubic morphology. Their examination, specifically in Li-ion batteries, demonstrated LNCO maintaining a minimal difference of 5 mA h g^{-1} , even on cycling from 2C to 0.1C. After 200 cycles, they recorded a specific capacity of 247 mA h g^{-1} , with a retention rate of 97.8 % (yielding an efficiency of 99.8 %) at 0.1C (Karupiah et al., 2020).

4.3. Olivines (olivine crystal structures/polyanion oxides)

An illustration of this concept is found in lithium iron phosphate (LFP), represented by the chemical formula LiFePO_4 . It is distinguished by a moderate capacity and a flat discharge plateau, typically ranging from 150 to 160 mA h g^{-1} (Yuan et al., 2010). Fig. 8a provides a schematic depiction of its crystal structure, while Fig. 8b shows the use of LiFePO_4 cathode in a cell. Within this structure, oxygen atoms arrange themselves in a hexagonal close-packed lattice. Phosphate anions fill tetrahedral voids, while iron and lithium cations occupy octahedral holes. The voids containing lithium cations create continuous, parallel channels running throughout the lattice (Stallard et al., 2022). However, under rapid insertion or extraction, lithium can also exist in a metastable solid solution of intermediate composition (Orikasa et al., 2013, De and Zhou, 2014).

These polyanion oxides are environmentally safe and demonstrate minimal capacity degradation over the battery's lifespan. In comparison to other cathode materials, these compounds undergo smaller volume changes during charge and discharge, as well as exhibit lower heat flow. This provides a significant safety advantage over cathodes based on Li-cobalt, making them preferable for more demanding applications. The transition metal iron (Fe) is not only cost-effective and readily available but also environmentally friendly (Hossain et al., 2023). However, olivines present a drawback in their lower energy density compared to other cathode chemistries like LiMn_2O_4 and $\text{LiNi}_{0.8}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$ for lithium batteries, as noted by Mauger and Julien (2018). Despite ongoing efforts to optimize LFP, recent focus has shifted towards enhancing the electrochemical properties of other isomorphous elements within the olivine family— LiMPO_4 , where M represents M = Ni, Mn, Co, or their combination. This emphasis aims to boost energy density relative to LiFePO_4 , leveraging the increasing redox potential $\text{M}^{3+}/\text{M}^{2+}$ versus Li^+/Li along this series (Mauger and Julien, 2018). Tolganbek et al. (2021) discuss high-voltage olivine-structured LiMPO_4 cathode materials for energy storage applications. They highlight olivine-structured cathodes such as LiCoPO_4 and LiNiPO_4 as promising candidates because of their operating potentials surpassing 5.0 V vs. Li^+/Li (Tolganbek et al., 2021).

However, despite their high voltage, these LiMPO_4 materials still face several intrinsic challenges that need to be addressed through various strategies before they can be widely used (Tolganbek et al., 2021). These issues and challenges include low electronic conductivity, poor Li^+ diffusion, a drop in charging voltage below 4 V, and limited stability of electrolytes at high voltage (Wu, 2019). To address these issues, two approaches have been suggested: the first involves reducing the size of the cathode particles via synthesis route variation, while the second entails using a nanocomposite of LiFePO_4 with a conductive carbon matrix. Carbon nanomaterials typically act as a conductive agent, enhancing electrical conductivity while also increasing material porosity, thus significantly reducing solid-state diffusion distances. These two mentioned approaches have paved the way for advancements in synthesizing olivines using other transition metals. Although synthesizing olivines with transition metals can be challenging and may result in limited capacity, they demonstrate high discharge potential and yield significant energy output (Eftekhari, 2017). To analyze and compare the various cathode materials reviewed, a summary is presented in Table 3 below, highlighting key characteristics such as electrochemical performance, stability, and cost-effectiveness across different materials.

5. Insights and future perspectives

As electric vehicle (EV) adoption continues to rise, the demand for higher energy-density batteries will become increasingly critical. Although current state-of-the-art Li-ion batteries, with energy densities ranging from 250 to 693 Wh L^{-1} and 100 to 265 Wh kg^{-1} , have significantly improved EV driving ranges, they still fall short compared to

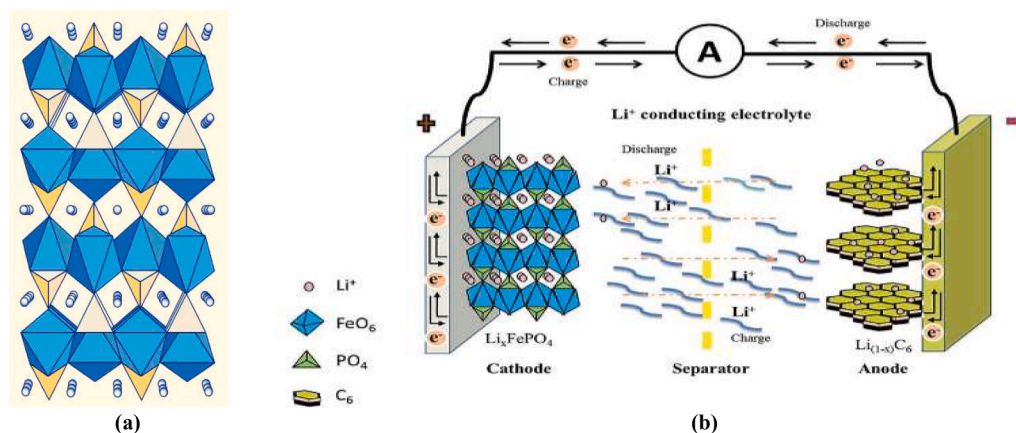


Fig. 8. a shows the olivine (LiFePO₄) crystal structure (Kurzweil and Brandt, 2019) and its schematic illustration in a typical Li-ion battery (Fig. 8b) (Cheng et al., 2016).

Table 3

Comparison of cathode materials for Li-ion batteries.

Cathode material	Performance metrics	Energy density (Wh/kg)	Cost	Safety	Applications	Citations
FeF ₂ , FeF ₃ (Conversion-type)	Theoretically ultrahigh energy densities	Up to 1500 Wh kg ⁻¹ (theoretical)	High (due to complex synthesis)	Capacity fading due to significant structural changes	Research phase, not widely used	(Nizam Uddin Khan et al., 2023, Moradi et al., 2023)
Solid-State Batteries	Higher energy density, longer lifespan, improved safety	300-500 Wh kg ⁻¹	High (current development stage)	Reduced risk of dendrite formation, better thermal stability	Next-gen EVs, portable electronics	(Moradi et al., 2023)
Lithium-Sulfur (Li-S) Batteries	High energy density, potential for cost reduction	Up to 600 Wh kg ⁻¹	Potentially lower than Li-ion	Faces challenges like polysulfide shuttle effect and capacity fading	Potential future for EVs, aerospace	(Zhang et al., 2018)
Nickel-Metal Hydride (Ni-MH)	Lower energy density, good cycle life, high discharge rate capability, good for deep discharge,	60-120	Low to Moderate	Considered safe, stable	Hybrid vehicles, stationary energy storage	(Ash et al., 2020, Fan et al., 2020)
Lithium-Air (Li-Air) Batteries	Very high theoretical energy density, electrocatalytic cathodic reactions; O ₂ interaction with carbon-based cathode	1700-2000 Wh kg ⁻¹	Very high (due to advanced materials and catalysts), (research stage)	Issues with oxygen reactivity, electrolyte stability, and scalability	Research phase, long-term potential for EVs	(Wang and Zhou, 2010, Capsoni et al., 2015, Imanishi and Yamamoto, 2019)
Magnesium-Ion (Mg-Ion) Batteries	Safer and potentially more energy-dense than Li-ion	400-500 Wh kg ⁻¹	High (due to development challenges)	Challenges with electrolyte and cathode material stability	Research phase, potential alternative to Li-ion	(Mao et al., 2019)
Lithium Cobalt Oxide (LCO)	High capacity (~140-150 m Ah g ⁻¹), fast charging, high electronic conductivity	(~540 Wh kg ⁻¹ theoretical)	High (cobalt is expensive)	Thermal instability, susceptible to thermal runaway, short lifespan	Portable electronics, smartphones, laptops, cameras	(Chen et al., 2024, Krieger et al., 2013)
Lithium Nickel Cobalt Aluminum Oxide (NCA)	Long service life (>2000 cycles), high specific energy~180-260 m Ah g ⁻¹ , Good specific power	Comparable to NMC, ~250-300 Wh kg ⁻¹	High (due to cobalt content)	Challenges with safety and cost, cobalt content raises safety concerns, recent advances to reduce cobalt and enhance thermal stability	EVs, energy storage systems, consumer electronics	(Zhang et al., 2024, Oh et al., 2004, Shana et al., 2020)
Lithium Iron Phosphate (LFP)	Long cycle life (>2000 cycles), stable voltage profile, low energy density, high power capability, lower voltage	90-160	Low	Very safe, high thermal and chemical stability	EVs, energy storage systems, stationary applications, grid stabilization	(Yuan et al., 2010, Stallard et al., 2022, Mauger and Julien, 2018, Tolganbek et al., 2021, Wu, 2019, Eftekhari, 2017)
Lithium Manganese Oxide (Li-Mn-O) Spinel	High thermal tolerance, good rate capability, flat voltage profile, capacity fading due to Mn ³⁺ instability and Jahn-Teller distortion	Varies, lower than LCO (~100-130)	Cost-effective (compared to LCO and NCA)	Safer, less toxic, minimal environmental impact, good thermal tolerance	High-discharge applications, power tools, hybrid vehicles, some portable electronics	(Huang et al., 2020, Zhang et al., 2000, Tomon et al., 2022, Callegari et al., 2022)
Lithium Nickel Manganese Cobalt (NMC)	High specific energy (~180-300 m Ah g ⁻¹), lifespan >1000 cycles, Issues with cation mixing and metal dissolution,	~900 Wh kg ⁻¹	Moderately high, but cheaper than cobalt-rich materials	Safer than cobalt-based cathodes, good thermal stability (a balance between stability and specific energy)	EVs, portable electronics, energy storage systems	(Jung et al., 2019, Tian et al., 2017, Shaju et al., 2002, Yabuuchi and Ohzuku, 2003, Yoshizawa and Ohzuku, 2007, Lee et al., 2021)

gasoline-powered vehicles. This gap underscores the need for ongoing innovation in battery technology to develop batteries with even higher energy densities, enabling EVs to travel greater distances on a single charge.

The future of cathode materials for Li-ion batteries is poised for significant advancements, driven by the need for not only higher energy densities but also improved safety and cost-effectiveness. Researchers are focusing on next-generation materials like high-voltage spinels and high-capacity layered Li-/Mn-rich oxides, alongside innovative composites that offer enhanced electrochemical performance and stability. Among these, lithium manganese oxide (Li-Mn-O) spinels stand out for their cost-effectiveness, non-toxicity, and high thermal tolerance, making them suitable for high-discharge applications such as power tools and hybrid vehicles. However, challenges such as capacity fading due to Mn^{3+} ion instability and structural transformations persist, prompting ongoing research into doping strategies and coatings to improve structural stability. For instance, doping Li-Mn-O spinels with elements like aluminum, magnesium, and fluorine has been shown to mitigate Mn^{3+} dissolution and enhance cycling stability, with aluminium doping in LiMn_2O_4 significantly reducing capacity fading by stabilizing the spinel structure. Similarly, surface coatings like Li_3PO_4 or Al_2O_3 create protective layers that reduce electrolyte decomposition and transition metal dissolution, further enhancing cycle life.

In addition to these developments, lithium metal oxides such as NMC and NCA oxides remain integral to EV applications due to their balance between energy storage capacity, durability, and cost-effectiveness. Yet, these materials face challenges including resource limitations, high costs, and safety concerns, which have spurred innovations like high-nickel variants, elemental doping (e.g., with zirconium), and advanced coating techniques. For example, high-nickel NMCs doped with zirconium demonstrate improved thermal stability and reduced capacity degradation, crucial for enhancing EV battery performance and cycle life. Nickel-rich materials like NMC 811 are particularly valued for their high energy densities and extended cycle lives, but future research will likely focus on reducing cobalt content, exploring further doping strategies for enhanced stability, and optimizing manufacturing processes to maximize energy density while minimizing costs. Given the environmental and ethical concerns surrounding cobalt mining, there is likely to be an increased emphasis on developing cobalt-free or cobalt-reduced cathode materials. In this context, manganese-rich cathodes could experience a resurgence, particularly in applications where safety and thermal stability are paramount, owing to manganese's ability to mitigate thermal runaway risks.

Moreover, integrating advancements in cathode materials with innovations in anode materials (e.g., silicon anodes) and electrolyte technologies (e.g., solid-state electrolytes) will be essential for achieving next-generation battery performance, which includes higher energy densities, faster charging, and longer lifespans. Beyond material innovations, the future of Li-ion batteries will also be shaped by advancements in battery management systems (BMS) and recycling technologies. Enhanced BMS will be critical for optimizing battery performance, safety, and lifespan, especially as EVs become more prevalent and battery packs grow larger and more complex. Meanwhile, as the demand for lithium and other critical materials continues to rise, effective recycling strategies will become increasingly important for ensuring a sustainable supply chain and reducing the environmental impact of battery production and disposal.

Overall, the future of Li-ion batteries is poised for significant advancements in performance, sustainability, and safety, which are essential for the widespread adoption of EVs and a sustainable energy transition. The rapidly evolving field of cathode materials will drive innovations in material development, performance optimization, and cost reduction, addressing the growing demands of EVs, grid storage, and portable electronics. These advancements could lead to EVs with extended ranges, reducing the need for frequent recharging and making electric transportation more accessible. As research progresses,

emerging battery technologies may bridge the current energy density gap, allowing EVs to compete with and surpass traditional internal combustion engine vehicles in range and performance.

6. Safety

Li-ion batteries, like other battery technologies, pose inherent hazards originating predominantly from both chemical and electrical sources, as well as risks arising from their processes. These batteries contain hazardous substances, such as lithium metal and flammable solvents, which, when exposed to temperatures exceeding a certain threshold, can trigger exothermic activity and uncontrolled reactions. Additionally, if Li-ion batteries are used beyond their safe operating parameters, they may result in the formation of lithium metal and thermal runaway (Lisbona and Snee, 2011). The chemicals used within the battery also pose potential environmental risks, which may occur due to spillage or the flammable nature of certain substances (Mrozik et al., 2021). The electrical hazard of Li-ion batteries is largely associated with their electrical energy content, which is determined by the state of charge and may lead to uncontrolled failure of the battery. Effective thermal management is crucial during intense charging and discharging processes to regulate the heat generated by electric current. Overcharging or discharging the battery can lead to unwanted exothermic reactions (Hafiz Muhammad, 2023). Batteries that undergo cyclical discharge and charging may experience an acceleration of temperature rise, leading to chemical instability in the battery materials/components. To mitigate this hazard, electronic protection based on voltage thresholds is often advised (Olabi et al., 2022). Historically, Li-ion batteries utilized in large-scale industrial applications exceeding 60 V have raised significant occupational health concerns. As a result, stringent safety measures have been put in place. All Li-ion batteries intended for use in high voltage applications (> 60 V) in large-scale industrial settings must adhere to recommended protection standards. These standards include insulation fault controls and terminals, which are designed to prevent potential exposure to hazardous battery conditions, such as electric shocks or thermal runaway (Ding et al., 2019). These measures are essential to ensure the safety of workers and to prevent accidents in industrial environments where high-voltage batteries are used.

Numerous safety tests, outlined by Chen et al. (2021), are primarily employed for EVs to mitigate or even prevent potential failures. However, a persistent concern revolves around how exposure to high temperatures in the battery may affect the thermal stability of the active materials (Zhao et al., 2017). Thermal runaways, often triggered by temperatures typically ranging from 130 °C to well over 200 °C, as well as internal short circuits, can result in ejection, combustion, explosion, and the release of toxic gases from a single battery, posing a significant risk (Ouyang et al., 2019). Conversely, cathode materials releasing O_2 at high temperatures (>150 °C) have been observed to exhibit high reaction rates and enthalpies, favouring inflammation, short-circuiting, and spontaneous battery combustion (Sharifi-Asl et al., 2019). These situations have the potential to escalate, leading to the propagation of thermal malfunction within a battery pack. To mitigate these risks, various measures are considered, such as implementing safety devices, incorporating fire-retardant additives, utilizing battery management systems (BMS), issuing hazard warnings, and being prepared for firefighting in the event of a hazard (Ouyang et al., 2019). Thus, ensuring safety requires a thorough risk assessment that scrutinizes and assesses various failure modes through fault tree analysis as an integral part of the manufacturing process (Zhao et al., 2024).

Additional ensembles of safety assessment and evaluation techniques/methods are discussed at length by Mohanty et al. (2024) and Peng and Jiang (2016). Cyclic voltammetry is employed to assess the electrochemical stability of the battery's components. Differential Scanning Calorimetry (DSC) is an analytical method that measures the thermal response of certain combinations of cell components across a

broad temperature range while keeping a constant scanning rate. This allows for the evaluation of the effects of thermal abuse on batteries (Mohanty et al., 2024). To further quantify and study the thermal stability of cells, Thermal Ramp Testing is employed, wherein the battery is heated until it fails in a linearly programmed manner. These tests, along with various others, help us understand the importance of adhering to safety regulations and standards set for Li-ion batteries (Peng and Jiang, 2016). Thanks to these and many more tests, we can better appreciate the significance of following safety guidelines, regulations, standards, and requirements for Li-ion batteries.

7. Conclusions

Understanding the roles and characteristics of key battery components, including anode and cathode materials, electrolytes, separators, and cell casing, is crucial for the development of advanced battery technologies, enhancing performance, safety, and sustainability. High-capacity, lightweight, and space-efficient batteries are in high demand and are expected to continue their steady rise with further technological advancements. Cathode materials play a pivotal role in the performance, safety, and sustainability of Li-ion batteries. This review examined the widespread utilization of various cathode materials, along with their respective benefits and drawbacks for specific applications. It delved into the electrochemical reactions underlying these battery technologies. Advances in these cathode materials continue to drive the development of safer, more efficient, and sustainable Li-ion batteries, with ongoing research focused on improving energy density, cycle life, and safety to meet the increasing demands of various applications. With ongoing research and technological advancements, Li-ion batteries are expected to continue their dominance in energy storage, particularly in EVs and plug-in hybrid EVs. Contemporary safety issues associated with Li-ion batteries, including thermal management and electronic protection, which are paramount in battery design and operation, have been addressed as well.

CRedit authorship contribution statement

Alex K. Koech: Writing – original draft, Conceptualization. **Ger-shom Mwandila:** Writing – review & editing, Supervision. **Francis Mulolani:** Writing – review & editing, Supervision. **Phenny Mwaanga:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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